



MOTIVATION

Despite Large Language Models (LLM) widespread application among diverse groups – students and corporate employees – current LLMs typically adopt a one-size-fits-all approach. (Fig 1)

This method **fails to consider individual user preferences and needs**, resulting in a significant limitation, diminishing the effectiveness of their responses. (Fig 1)

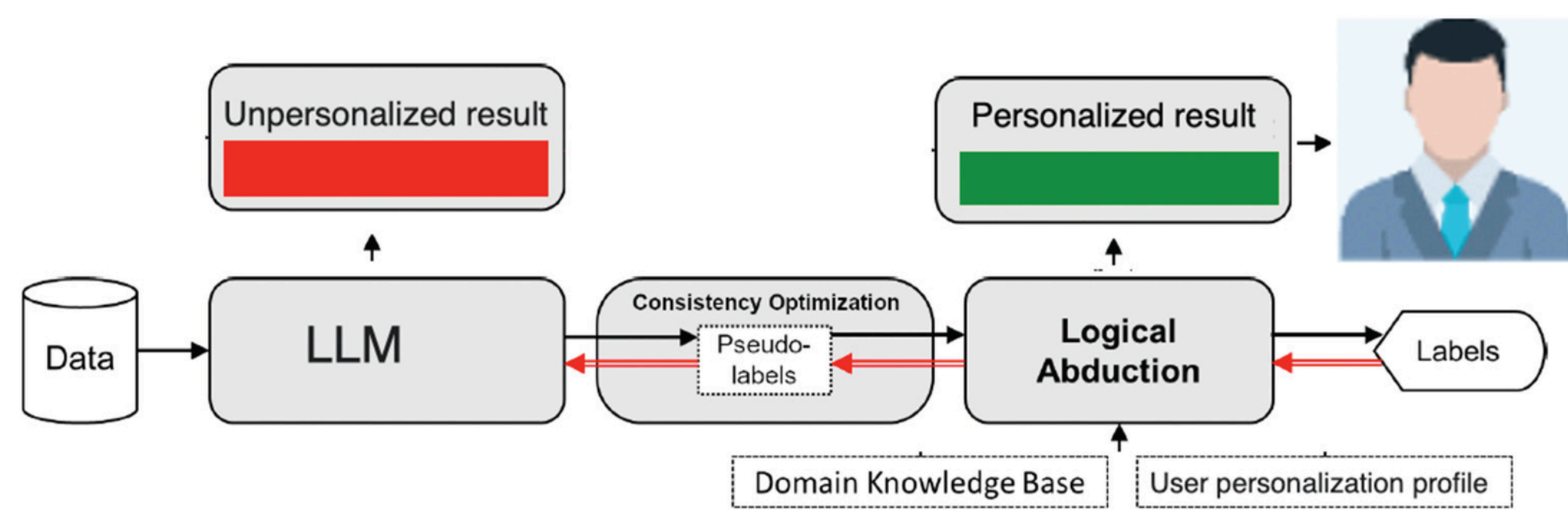


Fig 1: Bridging the Gap: Towards Personalized LLM Responses

RESEARCH QUESTION

How can we create and test a **question-answering system** to train **LLMs** to give a personalized response?

RESEARCH OBJECTIVE

Previous methods ensure consistent answers using user profiles but fail to handle new questions without relevant context.

Current issues:

- Depends on stored user data, thus ineffective for first-time queries.
- Struggles to adapt when no relevant context exists.
- Prioritize consistency of responses over flexibility.

Our goal is to collect **high-quality, novel data** on user preferences.

This will help develop a **personalized question-answering system** that tailors replies to individual needs, enhances engagement, and boosts LLM utility in education, work, and daily life.

User profiles will be created based on how the user engages with the system (as the model is given more data), and this is implemented using vLLM, Hugging Face, and python.

DATA COLLECTION

- **Platform:** Amazon Mechanical Turk (MTurk)
- **Participants:** Diverse global workforce
- **Demographics:** User's ChatGPT interactions
- **Questions:** 100 unique items on explicit preferences (e.g., tone, content format)
- **Interface:** Interactive JavaScript, HTML, CSS (Fig 2, Fig 3)
- **Ethics:** IRB approval, informed consent

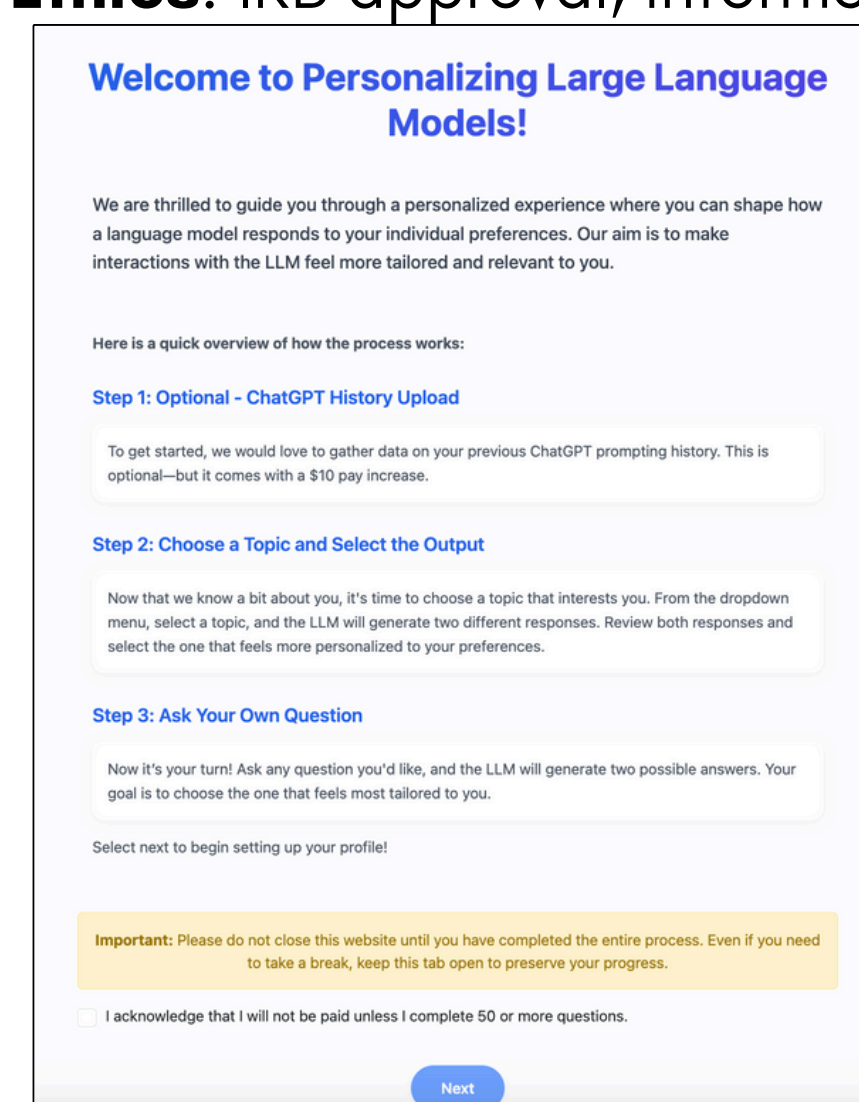


Fig 2: Landing Page

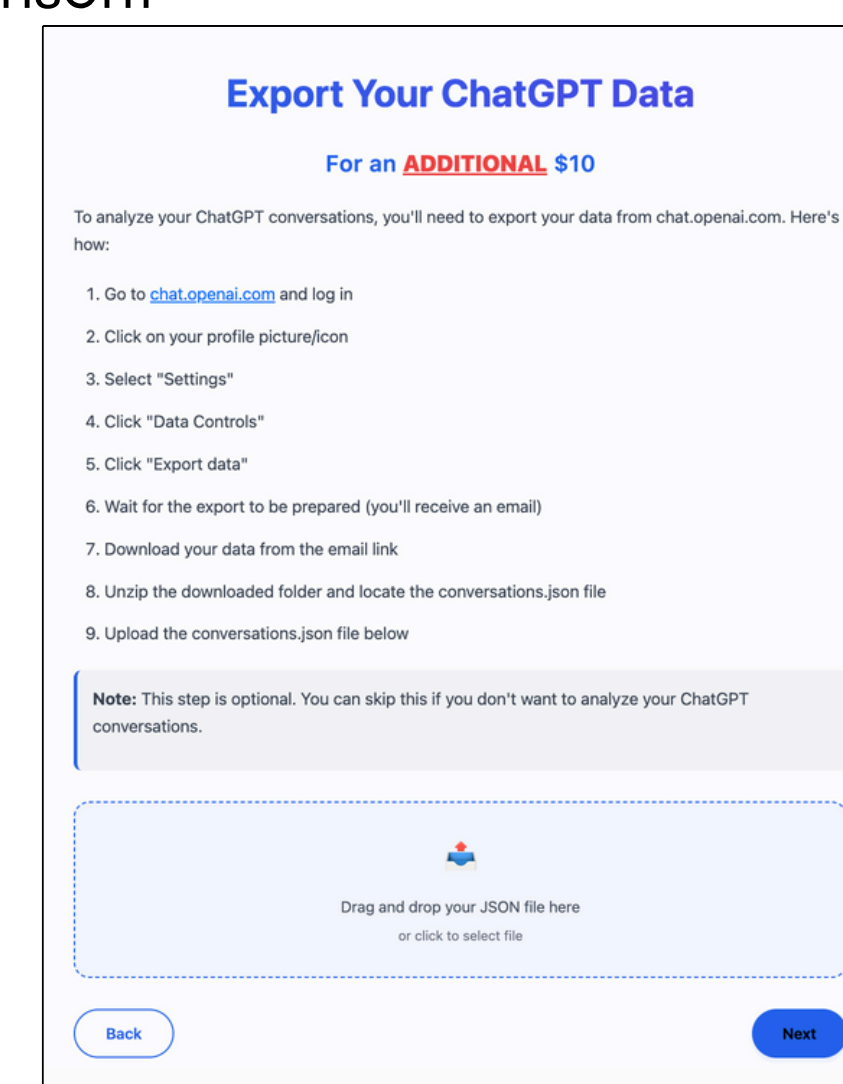


Fig 3: ChatGPT Export Page

METHODOLOGY

We created an interactive page where users select a category and question (Fig 4). The language model generates **two responses**, and users choose their **preferred** one. This process repeats for 50 unique questions.

Next, in Figure 5, users are asked to write a total of **50 unique questions**. The users write an explanation for their preferred response.

If the user is not satisfied with the LLM-generated responses, they have the option to generate new responses.

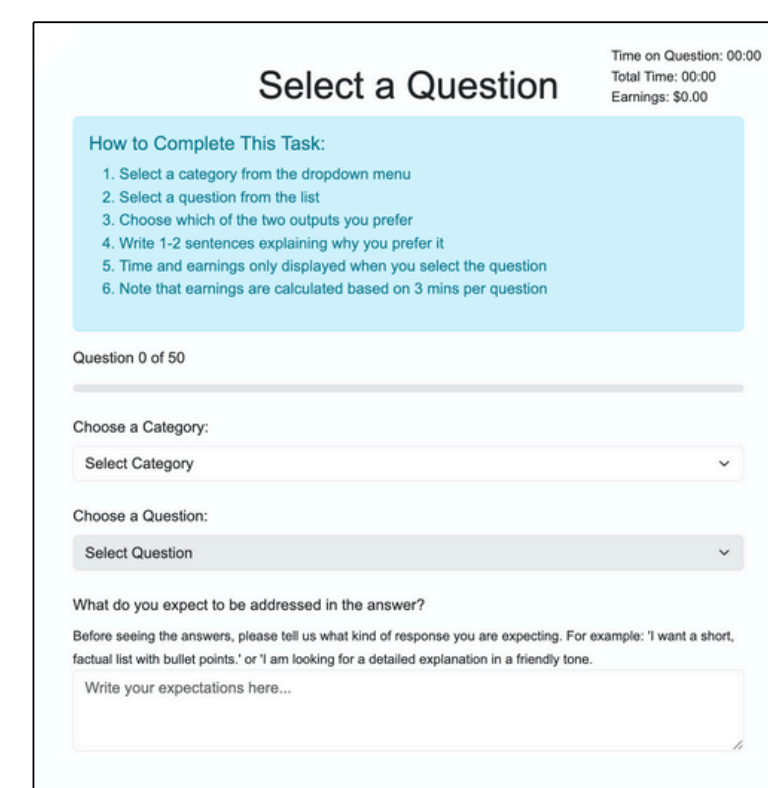


Fig 4: Selecting Question

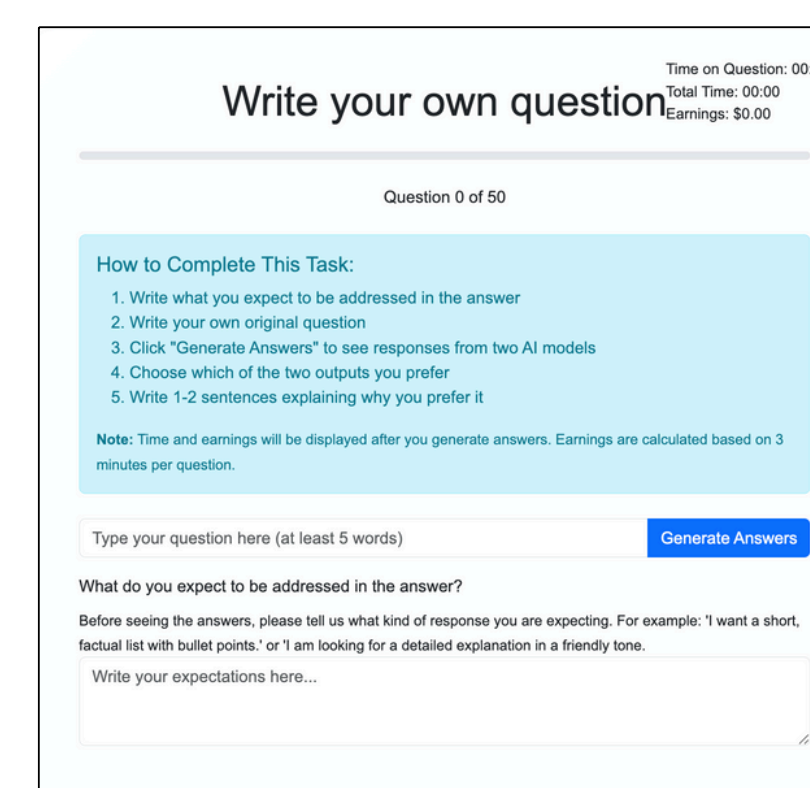


Fig 5: Create Own Question

RESULTS

Data Collected & Stored

- **User interaction logs:** ChatGPT history, questions selected, and both LLM responses stored in AWS DynamoDB
- **Preference tracking:** User-selected responses and their explanations recorded for model training
- **AWS infrastructure:** MTurk-integrated DynamoDB tables enable real-time preference data aggregation (Fig 6)
- **Training pipeline:** Curated datasets from user preferences feed continuous LLM improvement cycles

Name	Status	Partition key	Sort key	Indexes	Replication Regions	Deletion protection	Favorite	Read capacity mode	Write capacity mode	Total size	Table class
ChatHistory	Active	userid (S)	timestamp (N)	0	0	Off	☆	Provisioned (S)	Provisioned (S)	180.1 kilobytes	Standard
ProfileData	Active	userid (S)	-	0	0	Off	☆	Provisioned (S)	Provisioned (S)	4.7 kilobytes	Standard
Responses	Active	responseid (S)	-	1	0	Off	☆	Provisioned (S)	Provisioned (S)	4.8 megabytes	Standard
UserProgress	Active	userid (S)	-	0	0	Off	☆	On-demand	On-demand	102 bytes	Standard
Users	Active	userid (S)	-	0	0	Off	☆	Provisioned (S)	Provisioned (S)	73 bytes	Standard

Fig 6: Dataset Tables

FUTURE WORK

- **Deploy on MTurk:** collect 100-300 surveys from workers of different skillsets and backgrounds
- **Fine-tune LLMs** using collected survey data to adapt to individual user preferences
- **Test Personalization methods** using **Direct Preference Optimization** (Fig 7) to train the reward model

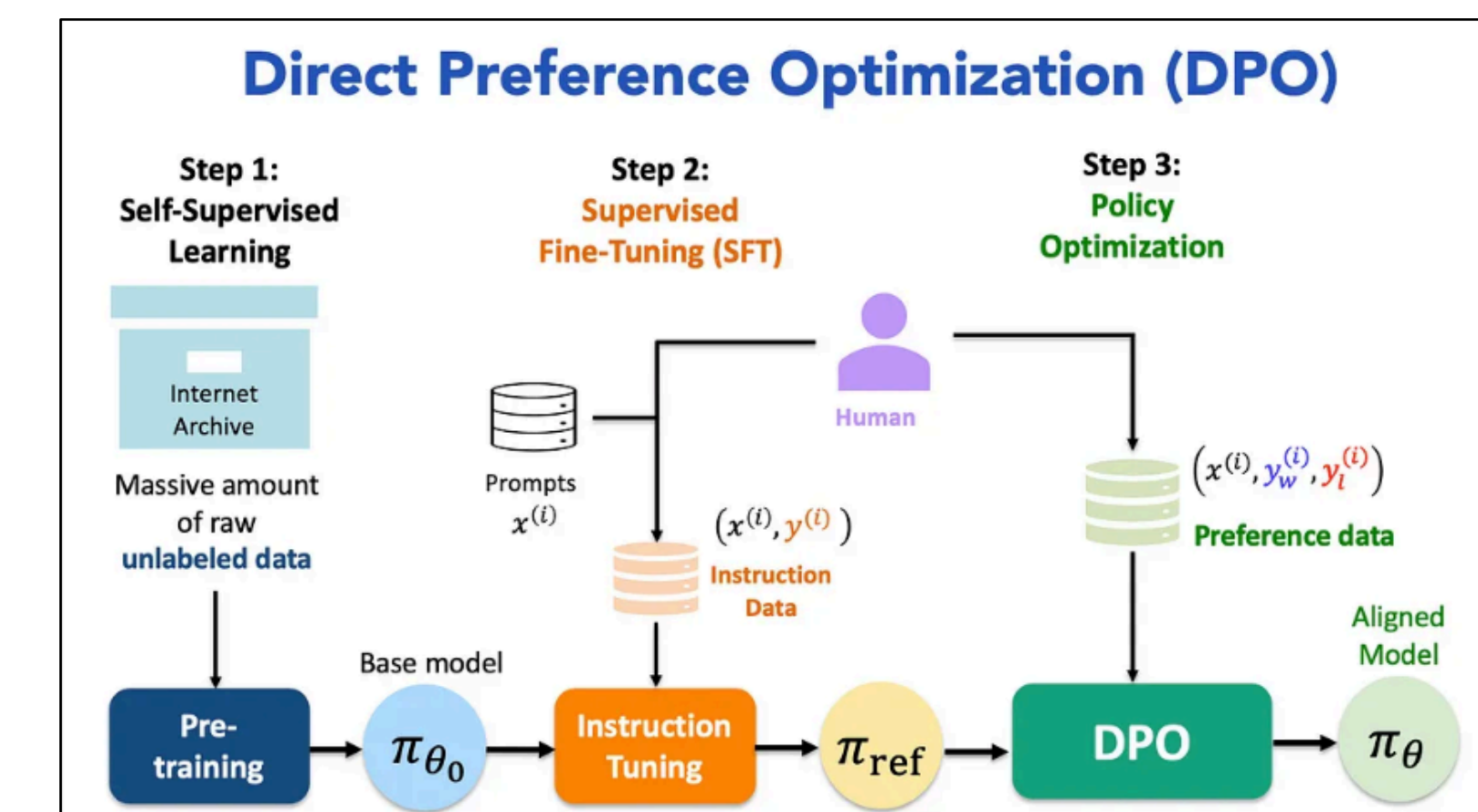


Fig 7: Training the reward Model

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